

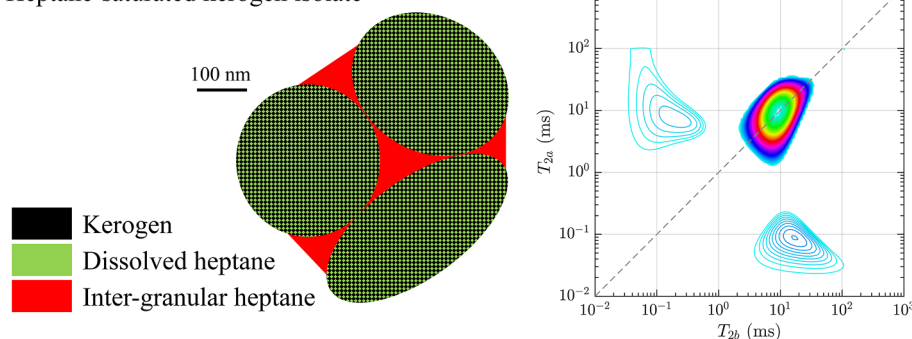
Diffusive coupling in heptane-saturated kerogen isolates evidenced by NMR T_1 - T_2 and T_2 - T_2 maps

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GRAPHICAL ABSTRACT

Heptane-saturated kerogen isolate



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ABSTRACT

It is well known that in unconventional shale, the ^1H NMR T_1/T_2 ratio for light producible hydrocarbons in the kerogen ($T_1/T_2 \approx 3 \leftrightarrow 6$) is larger than water in the clays ($T_1/T_2 \approx 2$). While such a contrast in T_1/T_2 is potentially useful for fluid typing and saturation in shale, the underlying mechanism remains elusive. Our previous NMR reports at $f_0 = 2.3$ MHz of heptane-saturated kerogen isolates distinguished two signals (a) inter-granular heptane between the kerogen grains with small $T_1/T_2 \approx 2.4$, and (b) dissolved heptane in the kerogen grain with large $T_1/T_2 \approx 70$. In this report we present new evidence of diffusive coupling effects between the inter-granular heptane and the dissolved heptane using T_1 - T_2 correlation maps as a function of saturation and T_2 - T_2 exchange maps. Our results indicate a phenomenon not previously reported; namely, different diffusive-coupling regimes between T_1 (with intermediate coupling) and T_2 (with weak coupling). We find that T_1/T_2 of the inter-granular heptane increases from $T_1/T_2 \approx 2.1 \rightarrow 6.5$ with decreasing saturation of the inter-granular pore-space, which we attribute to increasing confinement and the difference in diffusive coupling between T_1 and T_2 . Our findings imply that the large range $T_1/T_2 \approx 3 \leftrightarrow 6$ of the inter-granular (i.e. the more producible) hydrocarbons in shale is due to confinement together with diffusive-coupling effects with the dissolved (i.e. the less producible) hydrocarbons in kerogen.

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1. Introduction

Diffusive coupling is a well known phenomenon in porous media. It was first reported in water-saturated carbonates where macro-pore water was shown to be diffusively coupled to micro-pore water in micro-calcite grains [1]. Diffusive coupling was also shown to occur in water-saturated sandstones where macro-pore water couples to the clay flakes lining the macro-pores [2,3], and in water-saturated microporous glass beads [4]. One of the consequences of diffusive coupling is that in the strong diffusive-coupling regime, the NMR relaxation times $T_{1,2}$ for macro and micro-pore water merge into one peak, making them indistinguishable. Consequently, in the strong diffusive-coupling regime the $T_{1,2}$ distributions do not reflect the pore-size distribution. On the other hand, in the weak diffusive-coupling regime, the macro and micro-pore water have distinct $T_{1,2}$ peaks which can be interpreted as a pore-size distribution. Models have been proposed that predict the apparent surface relaxivity of the macro pores in the weak diffusive-coupling regime [1,3,5]. However these models do not explicitly account for pore-size or saturation of the macro-pores, although a semi-quantitative account of macro-pore saturation was reported in Ref. [1].

The difference in $T_{1,2}$ distributions for strong versus weak diffusively-coupled pores was recently reported for an organic-rich chalk [6,7]. In this study, bitumen blocks the diffusive coupling between macro-pore and micro-pore water, resulting in weak to intermediate diffusive-coupling. Meanwhile at shallower depths in the formation, the less mature organic-rich chalk has less bitumen and therefore no blocking mechanism between macro to micro-pore water, resulting in strong diffusive-coupling.

In the weak diffusive-coupling regime, the macro and micro-pore peaks are clearly separated in the T_2 distribution, and one can use T_2 - T_2 exchange maps to measure the exchange time between macro and micro-pores. As one increases the storage time in the T_2 -store- T_2 measurement, one observes the build up of off-diagonal peaks (i.e. exchange peaks) in the T_2 - T_2 map [8–13]. Exchange peaks in T_2 - T_2 maps have been observed in various systems including sandstones [8], cement pastes [9,11], and carbonates [14]. T_2 - T_2 exchange maps have also been used to study the effect of asphaltene deposition on pore-space morphology and wettability for sandstones [15].

More recently, exchange peaks have been observed in heptane-saturated organic-rich shales [16]. In this case, some of the exchange peaks in the T_2 - T_2 maps are postulated to be from magnetization transfer between the heptane (with longer T_2) and kerogen/bitumen (with shorter T_2). In other words, some of the exchange peaks are a result of exchange in nuclear magnetization rather than from diffusion of molecules between two pore-types. The results in Ref. [16] imply that ^1H - ^1H dipole-dipole relaxation dominates over surface paramagnetism in unconventional shales. Evidence for ^1H - ^1H dipole-dipole relaxation in kerogen was also reported in Ref. [17] by comparing T_1 - T_2 of water-saturated kerogen versus heavy-water saturated kerogen with $\text{H}_2\text{O}:\text{D}_2\text{O}$ proportions of 1:9, a techniques previously used in water-saturated carbonates [18]. Understanding which NMR relaxation mechanism dominates in shale is of fundamental interest when trying to understand the origin of the contrast in T_1/T_2 ratio, where T_1/T_2 is reported to be larger for light producible hydrocarbons ($T_1/T_2 \approx 3 \leftrightarrow 6$) compared to water in clays ($T_1/T_2 \approx 2$) [16,17,19–52].

We previously investigated the ^1H NMR relaxation mechanism in heptane-saturated kerogen isolates from an immature Kimmeridge shale outcrop, measured at a resonance frequency of $f_0 = 2.3\text{ MHz}$ [17,38,39]. The motivation for isolating the kerogen from the shale is that there is no interference from clay-bound water or hydroxyls [17,31,53,39,54]. We found two distinct features in the heptane-saturated kerogen (a) inter-granular heptane between the kerogen grains with long relaxation times $T_{2>} \approx 15\text{ ms}$ and small $(T_1/T_2)_{>} \approx 2.4$, and, (b) dissolved heptane in the kerogen with short relaxation times $T_{2\mu} \approx 0.24\text{ ms}$ and large $(T_1/T_2)_{\mu} \approx 70$. We postulate that the large $(T_1/T_2)_{\mu} \approx 70$ ratio of the dissolved peak is due to nano-confinement in

the organic pores of kerogen/bitumen. We tested this hypothesis on a polymer-heptane mix, where the high-viscosity polymer ($\approx 333,000\text{ cP}$) acted as a model for immature kerogen. Both NMR measurements [55–57] and molecular dynamics (MD) simulations [58] confirmed that the $T_{1s}/T_{2s} \approx 4$ for the surface relaxation of heptane was enhanced due to nano-confinement in the polymer matrix. These results strongly suggest that ^1H - ^1H dipole-dipole relaxation dominates over paramagnetism in organic nano-pores.

Characterizing the dissolved hydrocarbons in kerogen is also important for estimating the total gas in place in shale-gas reservoirs. Direct evidence of dissolved gas in kerogen comes from pressure-decay measurements of methane in shale, where it has been reported that dissolved gas constitutes $\approx 20\text{ wt\%}$ of the total gas [59], while the remaining gas is free ($\approx 40\text{ wt\%}$) and adsorbed ($\approx 40\text{ wt\%}$). Dissolved methane in shale has also been reported with in situ NMR measurements [40]. The presence of dissolved gas in kerogen could increase the reserve estimates in shale gas, and may be recoverable by pressure depletion, however it is generally not taken into account. The production of dissolved gas in kerogen could also be enhanced by CO_2 which preferentially dissolves in kerogen over methane [60].

In this report we extend our measurements of heptane-saturated kerogen isolates from an immature Kimmeridge shale outcrop. We show that the T_1/T_2 ratio of the inter-granular heptane increases from $T_1/T_2 \approx 2.1 \rightarrow 6.5$ as we de-saturate the inter-granular pore-space by controlled evaporation. We posit that the only explanation for an increase in T_1/T_2 with de-saturation is due to increasing confinement and a difference in diffusive coupling for T_1 (with intermediate coupling) versus T_2 (with weak coupling). To our knowledge, this is the first report of a difference in diffusive-coupling for T_1 versus T_2 . T_2 - T_2 exchange maps concur that T_2 is in the weak diffusive-coupling, from which we quantify the characteristic exchange time between the dissolved (i.e. the less producible) heptane and the inter-granular (i.e. the more producible) heptane. Our findings indicate that confinement together with diffusive-coupling play a crucial role in the large range $(T_1/T_2)_{>} \approx 3 \leftrightarrow 6$ previously reported for light producible hydrocarbons in shale [20,32].

2. Methodology

2.1. Kerogen preparation and characterization

The Kimmeridge shale originated from the coast of Dorset in the United Kingdom, and is the same as in Refs. [17,38,39,61]. LECO and RockEval measurements found a total organic carbon of $\text{TOC} \approx 10.1\text{ wt\%}$ (or $\approx 17.6\text{ vol\%}$ equivalently), a hydrogen index of $\text{HI} \approx 684\text{ mg-HC/g-TOC}$, an oxygen index of $\text{OI} \approx 12\text{ mg-CO}_2/\text{g-TOC}$, and a $T_{\text{max}} \approx 424^\circ\text{C}$, indicating Type II immature kerogen.

The isolation procedure was carried out at Stratum Reservoir as such:

1. Extraction of “free oil” (and some bitumen) with dichloromethane (DCM)
2. Acid digestion of calcite and dolomite with hydrochloric acid (HCl)
3. Multiple rinses with tap water
4. Acid digestion of silicate with hydrofluoric acid (HF)
5. Multiple rinses with tap water
6. Acid digestion with HCl
7. Multiple rinses with tap water
8. Sink-float separation with a ZnBr_2 solution of density 1.8 g/cm^3
9. Multiple rinses with tap water
10. Multiple rinses with deionized (DI) water
11. Acid digestion with HF
12. Multiple rinses with tap water
13. Multiple rinses with DI water
14. Freeze dry (low temperature dehydration)
15. Vacuum oven at 60°C



Fig. 1. Picture of kerogen pellets (top) pelletized at 77 kpsi stress; total mass of pellets is ≈ 1 g. SEM (scanning electron microscope) image of the crushed pellets after heptane saturation (bottom), taken from Ref. [17].

Inductively coupled plasma (ICP) of the isolates indicated a pyrite concentration of ≈ 8.2 wt% (or ≈ 3.5 vol% equivalently). We did not attempt to remove the pyrite using the chromous chloride method [62,63] in order to avoid contamination from paramagnetic Cr^{3+} ions on the kerogen surface. According to SEM (scanning electron microscopy) the pyrite was clustered in large $\sim 3 \mu\text{m}$ sized crystals [17], with no evidence of pyrite on the kerogen surfaces.

The kerogen powder was then further dried in a vacuum oven at 140°C at 0.3 bar for three days, then stored in a desiccator. The dry powder was then pelletized in an 8 mm diameter pellet press with an (axial) pelletizing stress of 26 kpsi, 51 kpsi, 77 kpsi, or 154 kpsi, where “kpsi” is short for 1000 psi. An example of the pellets is shown in Fig. 1 (top) at 77 kpsi, where the total dry mass was ≈ 1 g. The bulk density of the dry pellets $\rho_{b,0} = M_{\text{dry}}/BV_{\text{un-swollen}}$ was determined from the mass of the dry pellets M_{dry} , and the bulk volume of the un-swollen pellets before heptane saturation $BV_{\text{un-swollen}}$. The results listed in Table 1 indicate $\rho_{b,0} \approx 1.15 \text{ g/cm}^3$ at 51 kpsi, 77 kpsi, and 154 kpsi stress, and lower $\rho_{b,0} \approx 0.9 \text{ g/cm}^3$ at 26 kpsi stress. This indicates that a critical stress for compaction lies between 26 kpsi and 51 kpsi.

The pellets were then dried again in the vacuum oven, then NMR T_2 was measured on the dry pellets. As detailed in Ref. [39], the dry pellets had negligible NMR signal compared to background noise, indicating negligible moisture on the kerogen grains, and negligible signal from the kerogen or remaining bitumen. The pellets were then vacuum-saturated by applying a high vacuum for 2 h followed by saturation with *n*-heptane at 5000 psia hydrostatic pressure for 4 days.

Also included in this study are previous results after bitumen

extraction of the kerogen powder. The data in this report corresponds to “Extracted B” in Ref. [39], where extraction with toluene was carried out for 24 days before pelletizing and saturating with heptane.

The bulk density of the dry swollen pellets $\rho_b = M_{\text{dry}}/BV_{\text{swollen}}$ was determined from the mass of the dry pellets before heptane saturation M_{dry} , and the bulk volume of the swollen pellets after heptane saturation BV_{swollen} . The swelling factor δBV was then computed as such:

$$\delta BV = \frac{\rho_{b,0} - \rho_b}{\rho_b}. \quad (1)$$

As listed in Table 1, samples at 51 kpsi, 77 kpsi, and 154 kpsi pelletizing stress indicated large swelling factor of $\delta BV \approx 50\%$, which is a direct consequence of heptane dissolving in the kerogen grains [39,60]. The swelling was less $\delta BV \approx 40\%$ for the pellet formed at 26 kpsi stress, indicating a critical stress for compaction between 26 kpsi and 51 kpsi. The swelling was even less for the bitumen extracted sample $\delta BV \approx 23\%$, which is due to less dissolved heptane in the bitumen-extracted kerogen-grains (see below). We note that most of swelling took place along the axial direction of the pellet, i.e. in the direction of the axial stress during pelletization. We also note that the pellets kept the same bulk volume, i.e. they did not “de-swell”, after evaporation of heptane.

N_2 gas adsorption data at 77 K and BET (Brunauer, Emmett, Teller) analysis was performed on the pellets after completion of the heptane-saturated NMR measurements. The pellets were crushed into powder and dried at 110°C for 16 h before the gas-adsorption measurement of specific surface area S_{BET} . The inter-granular pore diameter $d_>$ was then determined using the following relation, which assumes spherical pores [39]:

$$\frac{\rho_b S_{\text{BET}}}{\phi_>} = \frac{6}{d_>}, \quad (2)$$

where $\phi_>$ is the inter-granular porosity according to NMR T_1 - T_2 (see below). Implicit in Eq. (2) is that S_{BET} only accounts for the surface area of the kerogen grain, in other words N_2 has limited solubility in kerogen at 77 K. Error analysis of the gas-adsorption data indicated that variation in S_{BET} between samples (Table 1) was within experimental uncertainties. As such, we use the average value $S_{\text{BET}} \approx 14 \text{ m}^2/\text{g}$ when computing $d_>$ in Eq. (2) for all the samples. As shown in Table 1, the pore diameter is $d_> \approx 200 \text{ nm}$, except of the 26 kpsi sample where $d_> \approx 300 \text{ nm}$, indicating a critical stress for compaction between 26 kpsi and 51 kpsi.

The grain diameter d_g is given by the following relation, which assumes spherical grains:

$$d_g = \frac{1 - \phi_>}{\phi_>} d_>. \quad (3)$$

We find that the average grain diameter for all the samples $d_g \approx 340 \text{ nm}$ is consistent with SEM images of the crushed pellets after heptane saturation shown in Fig. 1(bottom).

2.2. NMR measurement

The NMR data were acquired on an Oxford Instruments GeoSpec2 rock-core analyzer at a resonance frequency of $f_0 = 2.3 \text{ MHz}$ (similar to wireline NMR logging tools) using a customized 18 mm probe. The pellets were placed in a sealed glass tube, and left for one hour in the spectrometer to equilibrate to 30°C before data acquisition. The 2D T_1 - T_2 data were acquired with 32 log-spaced inversion recovery steps ranging from 0.1 to 1000 ms for T_1 encoding, followed by a CPMG (Carr, Purcell, Meiboom, Gill) train with an echo spacing of $t_E = 0.1 \text{ ms}$ for T_2 encoding. The 2D T_2 - T_2 data were acquired with 32 log-spaced decay times ranging from 0.1 to 500 ms for encoding T_{2a} , followed by storage time ranging from $t_\Delta = 0.1 \text{ ms}$ to 30 ms, followed by a CPMG train with $t_E = 0.1 \text{ ms}$ for encoding T_{2b} .

Table 1

Results of pore-size analysis including bulk density of dry un-swollen pellet ($\rho_{b,0}$), bulk density of dry swollen pellet (ρ_b), swelling of pellet bulk volume (δBV), inter-granular porosity from NMR ($\phi_>$), BET surface area (S_{BET}), inter-granular pore diameter ($d_>$), and grain diameter (d_g).

Sample	$\rho_{b,0}$ (g/cm ³)	ρ_b (g/cm ³)	δBV (%)	$\phi_>$ (pu)	S_{BET} (m ² /g)	$d_>$ (nm)	d_g (nm)
26 kpsi	0.90	0.64	40	53	17	360	320
51 kpsi	1.12	0.73	53	34	14	200	390
77 kpsi	1.14	0.78	47	37	13	210	350
154 kpsi	1.16	0.75	55	32	13	190	390
Extr. 77 kpsi	1.21	0.99	23	40	12	180	260

The 2D maps were processed using an inverse Laplace transform [64,65] with 120 log-spaced bins ranging from 10^{-2} ms to 10^3 ms for both T_1 and T_2 . Porosity is reported in “pu”, i.e. porosity units, or percent of bulk volume, equivalently, where the NMR hydrogen index of $HI = 1.0028$ for *n*-heptane is taken into account. The units of the relaxation distributions are in pu/div, i.e., pu per division, where for instance, for T_2 , “div” = $\log_{10}(T_{2,i+1}) - \log_{10}(T_{2,i})$ is independent of the bin index i . Saturation is reported in “su”, i.e. saturation units, or, percent of inter-granular pore-volume, equivalently. Note that while porosity and saturation are reported in “pu” and “su” units respectively, all the equations use porosity and saturation in fractions. The amplitude of the exchange peaks in the T_2 - T_2 map were determined by integrating the off-diagonal peaks using a polygon to encircle the exchange peaks, thereby limiting interference from the on-diagonal peaks.

There was no NMR signal on the scale of the plots for the dry kerogen pellets at $t_E = 0.1$ ms [39]. Increasing the echo-spacing from $t_E = 0.1$ ms to 0.2 ms did not alter the T_2 distribution of the heptane-saturated and partially-saturated pellets above $T_2 \geq 0.2$ ms, implying no internal magnetic-field gradient effects on T_2 [17].

At ambient conditions, *n*-heptane has a bulk relaxation time of $T_{1B} = T_{1B} = 1910$ ms (not de-oxygenated), a diffusion coefficient of $D_0 = 3.43 \mu\text{m}^2/\text{ms}$, a viscosity of $\eta = 0.390$ cP, and a density of $\rho_{C7} = 0.680$ g/cm³. All the measured relaxation times are shorter than ≤ 80 ms, implying that bulk relaxation is negligible and therefore $T_{1,2}$ are dominated by surface-relaxation. As such, the customary subscript S for $T_{1,2}$ indicating surface-relaxation is implied and omitted for clarity.

3. Results

3.1. T_1 - T_2 of fully saturated pellets

The T_1 - T_2 correlation maps of the heptane-saturated kerogen pellets are shown in Fig. 2 for different pelletizing stress, plotted as T_1 versus T_2 in (a) and T_1/T_2 versus T_2 in (b). The cutoff separating the inter-granular heptane from the dissolved heptane is chosen to be $T_2 = 1.7$ ms (dashed line). The inter-granular porosity $\phi_>$ above $T_2 > 1.7$ ms, the intra-granular porosity $\phi_<$ below $T_2 < 1.7$ ms, and the total NMR porosity ϕ_t (listed in the legend) are related by:

$$\phi_t = \phi_< + \phi_> \quad (4)$$

$$\phi_\mu = \frac{\phi_<}{1 - \phi_>}, \quad (5)$$

and ϕ_μ is the “micro-porosity”, i.e. the dissolved-heptane porosity of the kerogen grain. Fig. 3 (left) shows an illustration of three kerogen grains (black) of size $d_g \approx 340$ nm, dissolved heptane in the grains (green), and the inter-granular heptane (red).

Results of the dissolved heptane are listed in Table 2. We find that the dissolved heptane porosity is $\phi_\mu \approx 13$ pu at 51 kpsi, 77 kpsi, and 154 kpsi pelletizing stress. $\phi_\mu \approx 10$ pu is slightly lower at 26 kpsi, which may be due to experimental uncertainties. The bitumen extracted kerogen has significantly less dissolved heptane $\phi_\mu \approx 4$ pu, which indicates that heptane dissolves to a greater extent in bitumen than kerogen [39]. $T_{1\mu} \approx 16$ ms, $T_{2\mu} \approx 0.23$ ms, and $(T_1/T_2)_\mu \approx 70$ is found for all samples, indicating that the organic nano-pores in the kerogen/bitumen are the same size, as expected. The slight difference for 26 kpsi is likely due to uncertainties.

Results of the inter-granular heptane are listed in Table 3. We find that the inter-granular porosity is $\phi_> \approx 36$ pu at 51 kpsi, 77 kpsi, and 154 kpsi pelletizing stress, which is consistent with a random packing of mono-disperse spheres. $\phi_> \approx 53$ pu is higher at 26 kpsi, indicating a critical stress for compaction between 26 kpsi and 51 kpsi. $T_{1>} \approx 35$ ms, $T_{2>} \approx 15$ ms and $d_{>} \approx 200$ nm are the same at 51 kpsi, 77 kpsi, and 154 kpsi. $T_{1>} \approx 82$ ms and $T_{2>} \approx 39$ ms are larger for 26 kpsi, which is a result of larger pore-size $d_{>} \approx 300$ nm. $T_{1>} \approx 80$ ms and $T_{2>} \approx 35$ ms are larger for the bitumen extracted

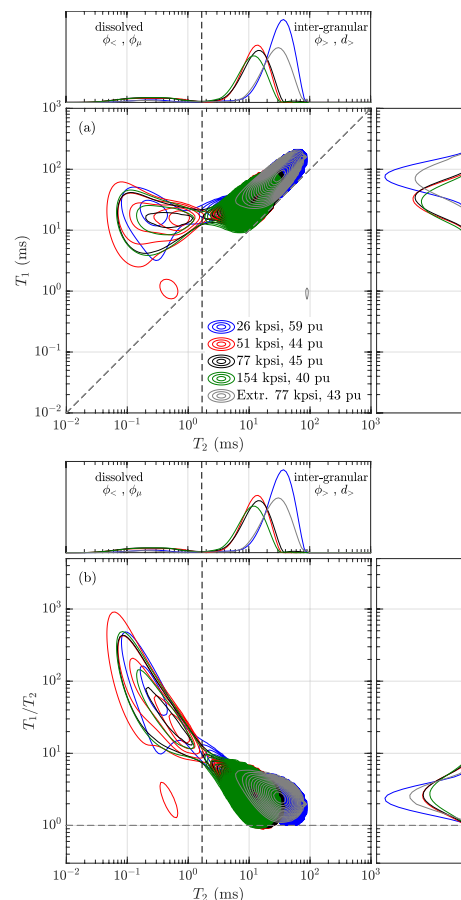


Fig. 2. (a) T_1 - T_2 correlation map for heptane-saturated kerogen pellets made using different pelletizing stresses. “Extr.” refers to kerogen with additional bitumen extraction using toluene. Subplots show 1D projections. Legend shows total NMR porosity ϕ_t in pu (i.e. porosity units). Cutoff at $T_2 = 1.7$ ms separates dissolved heptane from inter-granular heptane. Dry kerogen has negligible signal on the scale of this plot. (b) Same data plotted as a T_1/T_2 - T_2 correlation map.

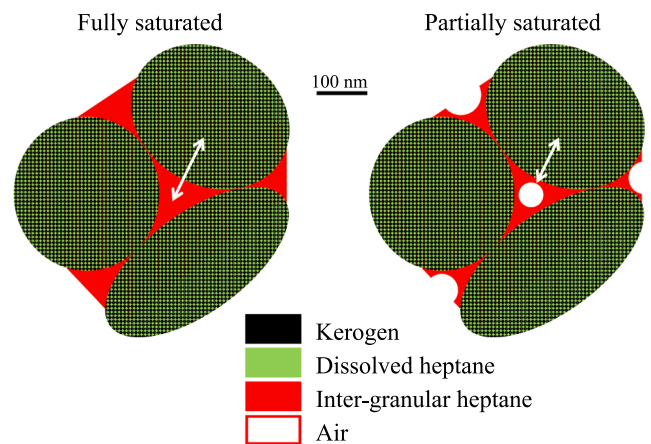


Fig. 3. Illustration of three kerogen grains (black), dissolved heptane in the kerogen (green), and inter-granular heptane (red) for fully saturated inter-granular pores (left), and partially-saturated inter-granular pores (right) with air (white) assuming heptane is the wetting fluid. Arrows illustrate the exchange between inter-granular and dissolved heptane.

sample, which is a result of less dissolved heptane (i.e. lower ϕ_μ) together with diffusive-coupling effects [39].

One can postulate from Fig. 2(a) that T_1 may be in the intermediate

Table 2

Results for dissolved heptane including intra-granular porosity $\phi_{<}$ taken below $T_2 < 1.7$ ms, and micro-porosity (i.e. amount of dissolved heptane) in the kerogen grain ϕ_{μ} , plus $T_{2\mu}$ and $(T_1/T_2)_{\mu}$ taken at the mode (i.e. peak) of the 2D map in Fig. 2, all as a function of pelletizing stress and for bitumen extracted “Extr.” kerogen.

Sample	$\phi_{<}$ (pu)	ϕ_{μ} (pu)	$T_{1\mu}$ (ms)	$T_{2\mu}$ (ms)	$(T_1/T_2)_{\mu}$
26 kpsi	7	10	21	0.21	104
51 kpsi	10	13	17	0.23	73
77 kpsi	8	12	15	0.23	67
154 kpsi	9	12	15	0.23	64
Extr. 77 kpsi	2	4	17	0.24	73

Table 3

Results for inter-granular heptane, including inter-granular porosity $\phi_{>}$ taken above $T_2 > 1.7$ ms, plus $T_{2>}$ and $(T_1/T_2)_{>}$ taken at the mode (i.e. peak) of the 2D map in Fig. 2, and the inter-granular pore-size $d_{>}$ (Eq. (3)), all as a function of pelletizing stress and for bitumen extracted “Extr.” kerogen.

Sample	$\phi_{>}$ (pu)	$T_{1>}$ (ms)	$T_{2>}$ (ms)	$(T_1/T_2)_{>}$	$d_{>}$ (nm)
26 kpsi	53	82	39	2.1	300
51 kpsi	34	35	14	2.4	200
77 kpsi	37	36	15	2.4	220
154 kpsi	32	31	13	2.5	200
Extr. 77 kpsi	40	80	35	2.4	200

diffusive-coupling regime due to the partial overlap between the inter-granular peak and the dissolved peak. One can also postulate from Fig. 2(a) that T_2 may be in the weak diffusive-coupling regimes due to the large separation of the inter-granular peak from the dissolved peak. While these observations from Fig. 2 are only postulates, i.e. they may be just coincidences, the increase in $(T_1/T_2)_{>}$ with de-saturation (see below) can only be explained by diffusive-coupling.

3.2. T_1 - T_2 of partially saturated pellets

The heptane-saturated kerogen pellets were de-saturated by controlled evaporation to air. This was achieved by opening the glass tube for a short period and allowing the heptane to partially evaporate. The effect of de-saturation on the 1D projections of the T_1 - T_2 maps are shown in Fig. 4 for the kerogen at 51 kpsi pelletizing stress. The saturation of the inter-granular pore-space is determined as such:

$$S_{>} = \frac{\phi_t(S_{>}) - \phi_{<}(1)}{\phi_t(1) - \phi_{<}(1)}, \quad (6)$$

where $\phi_t(1)$ and $\phi_{<}(1)$ refer to the total and intra-granular NMR porosities, respectively, at $S_{>} = 100$ su. The above definition for $S_{>}$ avoids having to use a cutoff at $T_2 = 1.7$ ms, which becomes inaccurate below $S_{>} \lesssim 30$ su as $T_{2>}$ approaches $T_2 = 1.7$ ms.

Fig. 4(a) and (b) clearly show that the distribution shifts to shorter T_1 and T_2 with decreasing $S_{>}$, respectively. Moreover, the shift in $T_{2>}$ (bullets) is more pronounced than $T_{1>}$ (bullets), which causes $(T_1/T_2)_{>}$ (bullets) to increase with decreasing $S_{>}$, as shown in Fig. 4(c). The increase in $(T_1/T_2)_{>}$ with decreasing $S_{>}$ poses a conundrum, which to the authors' knowledge has never been reported before. The conventional interpretation of surface relaxation in the fast-diffusion regime [66] is given by the following relation:

$$\frac{1}{T_{1>}} = \rho_{1>} \frac{6}{S_{>} d_{>}} \quad (7)$$

$$\frac{1}{T_{2>}} = \rho_{2>} \frac{6}{S_{>} d_{>}}, \quad (8)$$

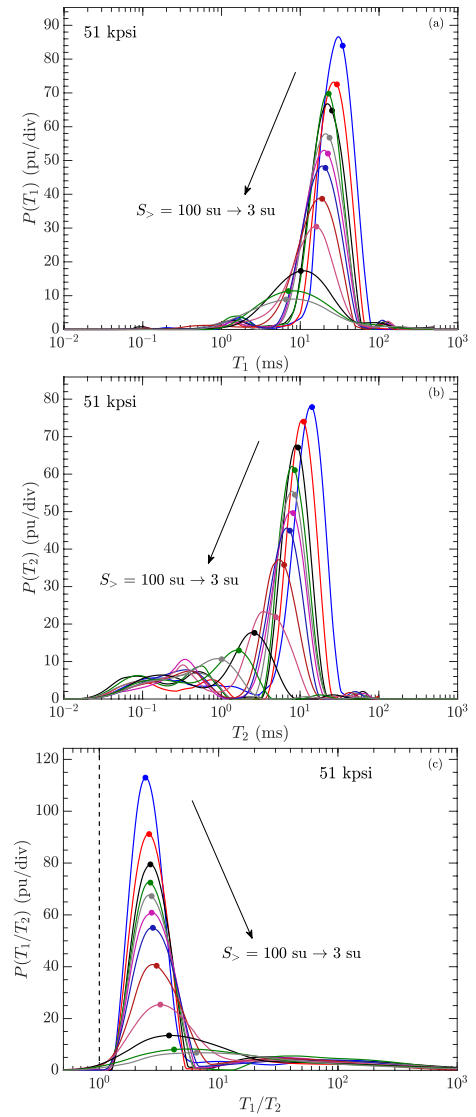


Fig. 4. (a) T_1 distributions (b) T_2 distributions and (c) T_1/T_2 distributions for heptane-saturated kerogen isolates pelletized at 51 kpsi stress, as a function of saturation of the inter-granular pore-space $S_{>}$ defined in Eq. (6). Bullets on the distributions correspond to modes (i.e. peaks) of the 2D T_1 - T_2 maps, defined as $T_{1>}$, $T_{2>}$ and $(T_1/T_2)_{>}$, respectively.

where the volume-to-surface ratio as a function of saturation is expressed as $S_{>} d_{>}/6$ since $d_{>}$ is computed assuming spherical pores (Eq. (2)). The surface-relaxivities $\rho_{1>}$ and $\rho_{2>}$ should not depend on saturation $S_{>}$, i.e. they are only functions of the fluid-surface interactions [66]. The resulting $(T_1/T_2)_{>}$ ratio should then be:

$$\left(\frac{T_1}{T_2}\right)_{>} = \frac{\rho_{2>}}{\rho_{1>}} \quad (9)$$

which should not depend on $S_{>}$. The only way for $(T_1/T_2)_{>}$ to depend on $S_{>}$ is for $\rho_{1>}$ and $\rho_{2>}$ to be (different) functions of $S_{>}$. We posit that the decrease in $S_{>}$ causes $T_{1>}$ to go from intermediate $\rightarrow$ strong diffusive-coupling, and $T_{2>}$ to go from weak $\rightarrow$ intermediate diffusive-coupling. Such changes in the extent of the diffusive-coupling cause $\rho_{1>}$ and $\rho_{2>}$ to depend on saturation $S_{>}$ (see below), and moreover for $\rho_{1>}$ and $\rho_{2>}$ to be different functions of $S_{>}$. We note that a semi-quantitative account of the effects of $S_{>}$ on $T_{2>}$ in Ref. [1] are in line with those reported here.

Fig. 5(a) shows the dependence of $(T_1/T_2)_{>}$ on $S_{>}$. All the samples show a monotonic increase in $(T_1/T_2)_{>}$ with decreasing $S_{>}$. The 51 kpsi

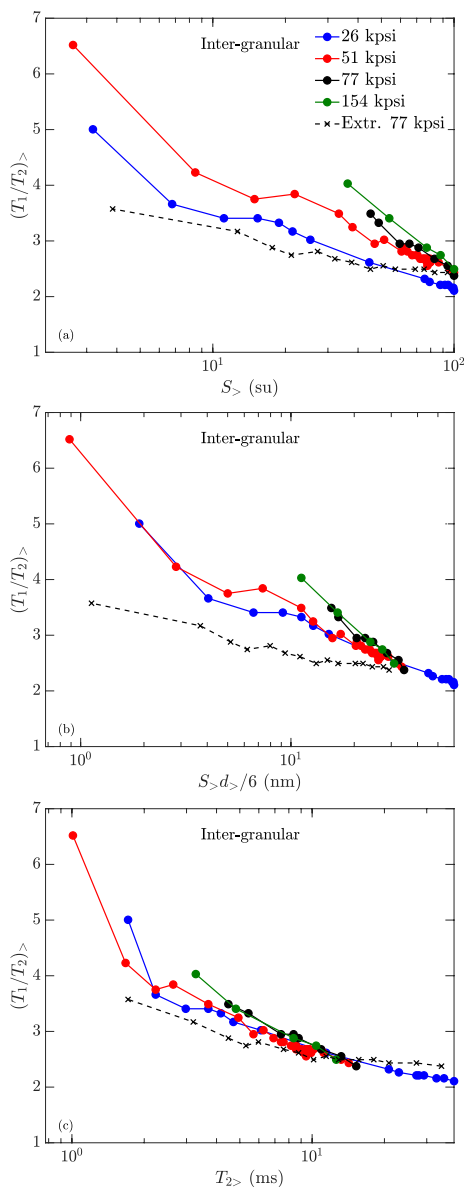


Fig. 5. $(T_1/T_2)_>$ of the inter-granular heptane in the kerogen pellets, as a function of inter-granular (a) saturation $S_>$ in su (i.e. saturation units), (b) volume-to-surface ratio as a function of saturation $S_>d_>/6$, and (c) $T_{2>}$. Data is shown for kerogen pellets made using different pelletizing stresses, and bitumen extracted “Extr.” kerogen.

sample, which has the most de-saturation data available, shows an increase from $(T_1/T_2)_> \approx 2.4 \rightarrow 6.5$ as saturation decreases from $S_> \approx 100 \rightarrow 3$ su. 77 kpsi and 154 kpsi agree with 51 kpsi over the limited range shown. 26 kpsi shows a similar trend, but with a lower $(T_1/T_2)_>$ at a given $S_>$. The bitumen extracted data has the least increase in $(T_1/T_2)_>$ with decreasing $S_>$, which corroborates with the weakest diffusive-coupling due to less dissolved heptane ($\phi_\mu \approx 4$ pu) [39].

As illustrated for the partially saturated case in Fig. 3 (right), in the presence of air the volume-to-surface ratio of the inter-granular space is $S_>d_>/6$, provided heptane remains the wetting fluid [15,67,68]. A large volume-to-surface ratio corresponds to the least confinement, while a small volume-to-surface ratio corresponds to the most confinement. Fig. 5(b) shows the dependence of $(T_1/T_2)_>$ on the volume-to-surface ratio $S_>d_>/6$ of the inter-granular space. Plotting against $S_>d_>/6$ causes the 26 kpsi data to collapse onto the 51 kpsi, 77 kpsi and 154 kpsi data, which suggests that the volume-to-surface ratio is

the underlying variable in $(T_1/T_2)_>$. In other words, decreasing the volume-to-surface ratio (i.e. increasing confinement) in the inter-granular pores enhances the diffusive-coupling effects for both $T_{1>}$ and $T_{2>}$. A similar trend is expected when decreasing the inter-granular pore-size $d_>$ with fixed $S_> = 100$ su, as previously been reported for $T_{2>}$ [3].

Finally, Fig. 5(c) shows the dependence of $(T_1/T_2)_>$ on $T_{2>}$, where all the 26 kpsi, 51 kpsi, 77 kpsi and 154 kpsi data collapse onto one curve. From Figs. 5(b) and (c) we can conclude that inter-granular pores with the least confinement have $T_{2>} \approx 40$ ms and $(T_1/T_2)_> \approx 2.1$, while inter-granular pores with the most confinement have $T_{2>} \approx 1$ ms and $(T_1/T_2)_> \approx 6.5$. This range is consistent with the range $(T_1/T_2)_> \approx 3 \leftrightarrow 6$ postulated for light producible hydrocarbons in shale [20,32]. We have therefore shown that the range $(T_1/T_2)_> \approx 3 \leftrightarrow 6$ for light hydrocarbons in shale is caused by the degree of confinement of the inter-granular (i.e. the more producible) pores, together with diffusive-coupling with dissolved (i.e. the less producible) hydrocarbons in the kerogen. Specifically, we find that nano-confinement of the order $S_>d_>/6 \approx 1$ nm causes the producible hydrocarbons to have large $(T_1/T_2)_> \approx 6.5$.

We note that while the above measurements and interpretation are for heptane/air saturated inter-granular pores, the heptane response should in principle be similar for heptane/water saturated inter-granular pores [1], provided heptane remains the wetting fluid [15,67,68]. In other words, these results are in principle applicable to hydrocarbon/water saturated shales.

3.3. $\rho_{1>}$ and $\rho_{2>}$ of partially saturated pellets

Fig. 6 shows the surface relaxivities $\rho_{1>}$ and $\rho_{2>}$ determined from Eqs. (7) and (8), respectively. Above $S_> \gtrsim 30$ su, $\rho_{1>}$ is roughly constant while $\rho_{2>}$ increases with decreasing $S_>$, which causes $(T_1/T_2)_>$

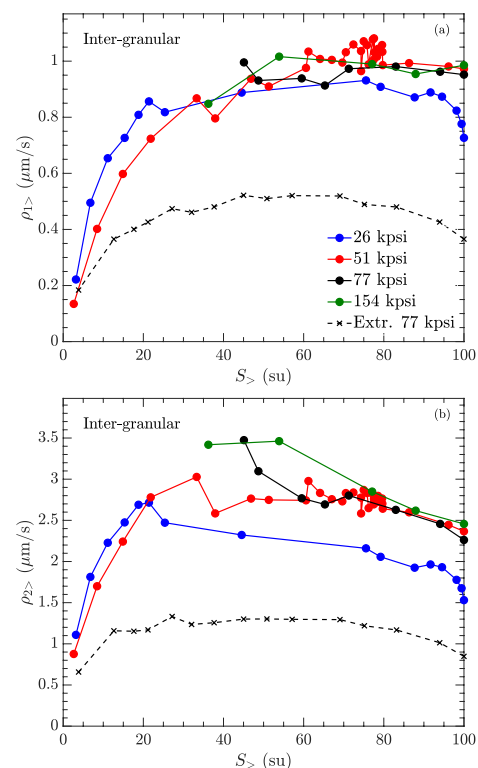


Fig. 6. Surface relaxivities (a) $\rho_{1>}$ and (b) $\rho_{2>}$, defined in Eqs. (7) and (8) respectively, of the inter-granular heptane in the kerogen pellets, as a function of inter-granular saturation $S_>$ in su (i.e. saturation units). Data is shown for kerogen pellets made using different pelletizing stresses, and bitumen extracted “Extr.” kerogen.

(Eq. (9)) to increase with decreasing $S_{>}$. Below $S_{>} \lesssim 30$ su, both $\rho_{1>}$ and $\rho_{2>}$ decrease with decreasing $S_{>}$. The decrease below $S_{>} \lesssim 30$ su is more pronounced for $\rho_{1>}$ than for $\rho_{2>}$, which causes $(T_1/T_2)_{>}$ to increase with decreasing $S_{>}$. In other words, if the extent of the decrease for $\rho_{1>}$ and $\rho_{2>}$ were the same, $(T_1/T_2)_{>}$ would be constant. Note that the cross-over at $S_{>} \simeq 30$ su corresponds to a volume-to-surface ratio of $S_{>} d_{>}/6 \simeq 10$ nm, see Fig. 5(b).

An expression for the “apparent” surface-relativity ρ_{2app} (which is equivalent to $\rho_{2>}$ in this report) has previously been proposed in the weak diffusive-coupling regime [1,3,5]. However, the apparent relativity theory for $\rho_{2>}$ does not take pore-size or saturation of the intergranular pore space into account, nor does it work in the intermediate or strong diffusive-coupling regimes. We note however that the apparent relativity theory does account for the lower $\rho_{1>}$ and $\rho_{2>}$ for the bitumen extracted sample at $S_{>} = 100$ su in Fig. 6, which is a result of less dissolved heptane $\phi_{\mu} \simeq 4$ pu (Table 2) in kerogen [39]. The slightly lower $\rho_{1>}$ and $\rho_{2>}$ for the 26 kpsi sample at $S_{>} = 100$ su could also be due to less dissolved heptane $\phi_{\mu} \simeq 10$ pu (Table 2), or could be from experimental uncertainties.

We also note that according to the apparent relativity theory, for a given ϕ_{μ} , a lower surface-relativity leads to stronger diffusive-coupling [1,3,5]. This is consistent with our interpretation of Fig. 6 where the decrease in $\rho_{1>}$ and $\rho_{2>}$ below $S_{>} \lesssim 30$ su is a result of stronger diffusive-coupling due to increasing confinement in the inter-granular pore-space.

We note in passing that the fast-diffusion limit [66] applies throughout, i.e. $\rho_{1>} d_{>}/D_0 \ll 1$ and $\rho_{2>} d_{>}/D_0 \ll 1$, which validates the use of Eqs. (7) and (8), respectively.

3.4. T_2 - T_2 of fully saturated pellet

T_2 -store- T_2 is a powerful technique for determining the exchange time τ_{ex} between diffusively coupled macro and micro-pores [8–13,16], or in our case inter-granular and dissolved heptane, respectively. The first requirement for observing the exchange peaks and determining τ_{ex} is weak diffusive-coupling since the inter-granular and dissolved heptane must have distinct (i.e. well separated) T_2 peaks. The second requirement is that the exchange time τ_{ex} must be comparable to or less than the average (i.e. log-mean) T_{1LM} , i.e. $\tau_{ex} \lesssim T_{1LM}$. This condition arises because there is T_1 decay during the storage time t_{Δ} in the T_2 -store- T_2 measurement, and the signal cannot decay to the noise level before exchange is observable.

In Fig. 7 we see that both of these conditions are fulfilled. At the shortest storage time of $t_{\Delta} = 0.1$ ms in Fig. 7(a), we see two distinct peaks on the diagonal, with a hint of an exchange peak. At a longer storage time of $t_{\Delta} = 20$ ms in Fig. 7(b), we clearly see the exchange peaks symmetrically distributed off-diagonally.

In Fig. 8(a) we plot the total NMR porosity ϕ_t as a function of t_{Δ} , which (in part) decays as a result of T_1 decay during storage [13]. Meanwhile, the integral of the NMR exchange peaks ϕ_{ex} increases initially up to $t_{\Delta} \simeq \tau_{ex,ef} \simeq 9$ ms, then decays due to T_1 decay during storage. The dissolved peak in Fig. 7(b) disappears from the on-diagonal location at $T_{2\mu} \simeq 0.23$ ms, which according to Ref. [12] is a direct consequence of T_1 decay during storage.

In Fig. 8(b) we plot the ratio ϕ_{ex}/ϕ_t as a function of t_{Δ} , and we find an exponential-like recovery. A simplified expression to the t_{Δ} dependence in ϕ_{ex}/ϕ_t is given by [13]:

$$\frac{\phi_{ex}}{\phi_t} = A \left[1 - \exp\left(-\frac{t_{\Delta}}{\tau_{ex,ef}}\right) \right] \quad (10)$$

$$\frac{1}{\tau_{ex,ef}} = \frac{1}{T_{1LM}} + \frac{1}{\tau_{ex}}. \quad (11)$$

The two free parameters in the fit are the effective exchange time $\tau_{ex,ef} \simeq 9$ ms and asymptotic fraction of the exchange-peak porosity

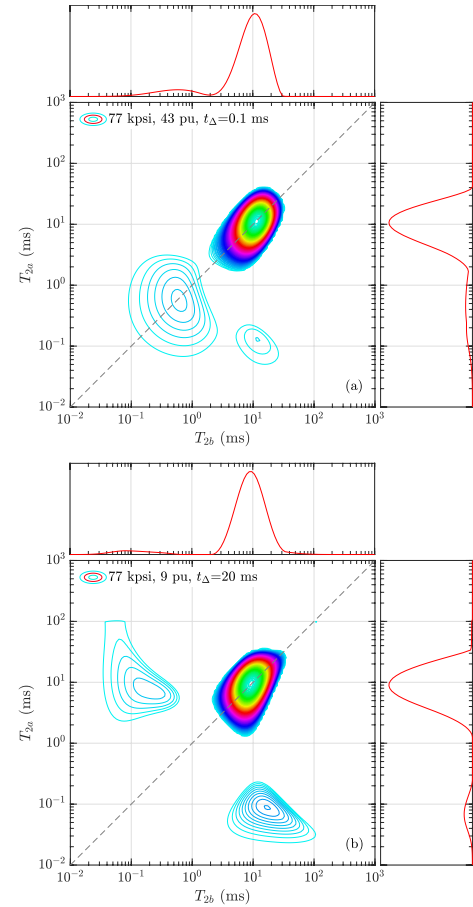


Fig. 7. T_2 - T_2 exchange maps for heptane-saturated kerogen isolates pelletized at 77 kpsi stress, using a storage time of (a) $t_{\Delta} = 0.1$ ms, and (b) $t_{\Delta} = 20$ ms, all fully saturated ($S_{>} = 100$ su). Subplots show 1D projections. Legend shows total NMR porosity ϕ_t in pu (i.e. porosity units). Dry kerogen has negligible signal on the scale of this plot.

$A \simeq 13\%$, which are determined using non-linear least squares. According to Eq. (11) [8], the intrinsic exchange time $\tau_{ex} = 15$ ms can then be derived from $\tau_{ex,ef} \simeq 9$ ms and $T_{1LM} \simeq 27$ ms.

An estimate of the strength in diffusive-coupling can then be estimated from the following expressions [14]:

$$C_1 = \frac{T_{1\mu}}{\tau_{ex}} \simeq \frac{16 \text{ ms}}{15 \text{ ms}} \simeq 1 \quad (\text{intermediate}) \quad (12)$$

$$C_2 = \frac{T_{2\mu}}{\tau_{ex}} \simeq \frac{0.23 \text{ ms}}{15 \text{ ms}} \simeq 0.02 \quad (\text{weak}) \quad (13)$$

where C_1 is the strength in diffusive coupling for T_1 , and C_2 is the strength in diffusive coupling for T_2 . According to Ref. [14], $C_{1,2} < 0.1$ indicates weak diffusive-coupling, and $C_{1,2} > 10$ indicates strong diffusive-coupling, with intermediate diffusive-coupling for $0.1 < C_{1,2} < 10$. Eqs. (12) and (13) therefore indicate intermediate diffusive-coupling for T_1 , and weak diffusive-coupling for T_2 . This is in agreement with the qualitative observations from Fig. 2(a) where inter-granular and dissolved heptane have partially overlapping peaks in T_1 (i.e. intermediate coupling), and distinct peaks in T_2 (i.e. weak coupling).

Another interpretation for the exchange peaks in T_2 - T_2 maps is magnetization transfer between kerogen/bitumen (with short T_2) and heptane (with long T_2) [16]. We note however that in Ref. [16], the short T_2 peak from kerogen/bitumen is present before heptane-saturation, i.e. it is present in the “as-received” shale. In our kerogen isolates, there is no NMR signal in the T_1 - T_2 or T_2 - T_2 maps before heptane

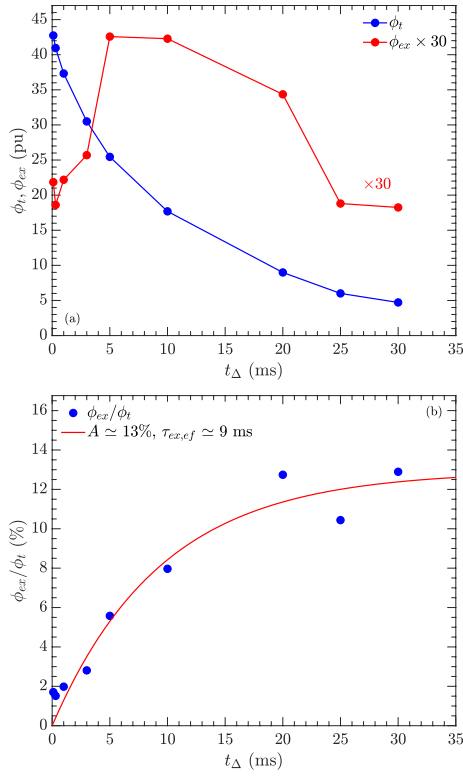


Fig. 8. (a) Total NMR porosity (ϕ_t), and porosity of off-diagonal NMR exchange peaks (ϕ_{ex}) multiplied by $\times 30$ for clarity, as a function of storage time t_Δ . (b) ϕ_{ex}/ϕ_t ratio (bullets) as a function of storage time t_Δ , and fit using Eq. (10) (curve).

saturation, implying that the short T_2 peak in Figs. 2 and 7 is heptane dissolved in kerogen. We also note that in our heptane-saturated kerogen isolates, there is no ambiguity about which fluid (i.e. water or hydrocarbon) is saturating which pore-type (i.e. organic or inorganic).

From mass balance [17], we can also conclude that the short T_2 peak in Figs. 2 and 7 is not bitumen which becomes visible (i.e. its T_2 becomes detectable) because its viscosity is lowered by dissolved heptane. As such, our interpretation of the exchange peaks in Fig. 7 is diffusive coupling, and not magnetization transfer.

As a consistency check, we calculate the penetration depth $L_{D,ex}$, or restricted diffusion [31,69] length, of heptane in kerogen during exchange time $\tau_{ex} = 15 \text{ ms}$:

$$L_{D,ex} = \sqrt{D_0 \tau_{ex} \phi_\mu^{m-1}} \approx 65 \text{ nm} \approx d_g/5 \quad (14)$$

where $m \approx 5.6$ is the Archie cementation exponent of dissolved heptane in kerogen [39]. We note that in Ref. [39] a cementation exponent of $m \approx 6.3$ was derived using the log-means of T_1 and T_2 , however using the mode (i.e. peak) of the 2D T_1 - T_2 instead results in $m \approx 5.6$.

For a spherical grain of diameter $d_g \approx 340 \text{ nm}$, Eq. (14) suggests that heptane penetrates $L_{D,ex} \approx 65 \text{ nm}$ into the grain surface during time τ_{ex} , which corresponds to $(d_g^3 - (d_g - 2L_{D,ex})^3)/d_g^3 \approx 75\%$ of the grain volume. In other words, $\approx 75\%$ of the kerogen grain, i.e. the majority of the kerogen, is penetrated during $\tau_{ex} \approx 15 \text{ ms}$, which is reasonable for the definition of an exchange time. The arrow in Fig. 3 (left) illustrates the exchange between inter-granular and dissolved heptane, and the penetration depth $L_{D,ex} \approx 65 \text{ nm}$ (i.e. the length of the arrow) into the kerogen grain during time τ_{ex} .

A similar analysis to Eq. (14) indicates that the penetration depth during time $T_{1\mu} \approx 16 \text{ ms}$ is given by:

$$L_{D,1} = \sqrt{D_0 T_{1\mu} \phi_\mu^{m-1}} \approx 68 \text{ nm} \quad (15)$$

$$L_{D,2} = \sqrt{D_0 T_{2\mu} \phi_\mu^{m-1}} \approx 8 \text{ nm} \quad (16)$$

In other words, $\approx 77\%$ of the kerogen grain is penetrated during $T_{1\mu}$, which is reasonable for intermediate diffusive-coupling for T_1 . On the other hand, the penetration depth during time $T_{2\mu} \approx 0.23 \text{ ms}$ is small $L_{D,2} \approx 8 \text{ nm}$, where only $\approx 13\%$ of the kerogen grain is penetrated, which is consistent with weak diffusive-coupling for T_2 .

Fig. 3 (right) illustrates the effect of partial saturation of the inter-granular pore-space, provided heptane is the wetting fluid. The exchange time τ_{ex} between inter-granular and dissolved state will shorten with decreasing $S_>$ since the time needed to diffuse from the center of the inter-granular pore to the grain surface will shorten (see shortening of arrows in the inter-granular pore). According to Eqs. (12) and (13), a shorter τ_{ex} leads to stronger diffusive-coupling for both T_1 and T_2 . This trend is consistent with the decrease in $\rho_{1>}$ and $\rho_{2>}$ with decreasing $S_>$ in Fig. 6, which is also interpreted as due to stronger diffusive-coupling.

3.5. Effects of reservoir conditions

All of the above data were acquired at laboratory conditions of 30°C and 1 atmosphere. It is worth postulating what the above results would be under reservoir conditions, as measured by an NMR logging tool downhole. Reservoir conditions will typically involve higher temperature, higher pressures, and multiple components in the reservoir oil. In the case of shale oil with a high gas-to-oil ratio, the reservoir oil will have components lighter than heptane such as methane and natural gas liquids.

It is therefore reasonable to expect that reservoir conditions will decrease viscosity η and increase the bulk diffusion coefficient D_0 of the producible oil, given that $D_0 \propto (\eta/T)^{-1}$ [70]. Given the relation $L_{D,ex} \approx d_g/5$ (Eq. (14)), this will tend to decrease the exchange time $\tau_{ex} \propto 1/D_0$ between the dissolved and inter-granular fluid, resulting in larger $C_1 \propto D_0$ and $C_2 \propto D_0$ coefficients (Eqs. (12) and (13)), and therefore stronger diffusive coupling for both T_1 and T_2 . By analogy with viscous fluids [57], we also postulate that $T_{1\mu}$ of the dissolved hydrocarbon is roughly independent of η/T and D_0 , while $T_{2\mu} \propto (\eta/T)^{-1/2} \propto D_0^{1/2}$, resulting in $C_1 \propto D_0$ and $C_2 \propto D_0^{3/2}$. In other words, reservoir conditions will result in stronger diffusive-coupling for both T_1 and T_2 , thereby increasing $T_{1>}$ and $T_{2>}$ for inter-granular pores, perhaps more so for $T_{2>}$.

Another effect of reservoir conditions is that the reservoir oil will have components lighter than heptane, which will decrease the dissolved porosity ϕ_μ since lighter components have lower solubility in kerogen [60]. One also expects lower ϕ_μ with increasing kerogen maturity [60]. The effect of lower ϕ_μ will be similar to the bitumen extracted results, although probably not as severe.

4. Conclusions

We present two evidence for diffusive-coupling effects between inter-granular heptane and heptane dissolved in the isolated kerogen grains of immature Type II kerogen from the Kimmeridge. The first evidence is from T_1 - T_2 correlation maps as a function of saturation, and the second is from T_2 - T_2 exchange maps.

From T_1 - T_2 correlation maps we find that $(T_1/T_2)_>$ for inter-granular heptane increases from $(T_1/T_2)_> \approx 2.1 \rightarrow 6.5$ with decreasing saturation $S_> = 100 \rightarrow 3 \text{ su}$ (saturation units) of the inter-granular pore-space. To the authors' knowledge, this phenomenon has never been reported before, and can only be explained if T_1 and T_2 are in different diffusive-coupling regimes; namely, if T_1 is in the intermediate diffusive-coupling regime while T_2 is in the weak diffusive-coupling regime.

Using the volume-to-surface ratio as a function of saturation of the inter-granular pores ($S_> d_>/6$), we show that inter-granular pores with $S_> d_>/6 \approx 60 \text{ nm}$ (i.e. the least confinement) have small $(T_1/T_2)_> \approx 2.1$, while inter-granular pores with $S_> d_>/6 \approx 1 \text{ nm}$ (i.e.

the most confinement) have large $(T_1/T_2)_> \approx 6.5$. Similar behavior is expected when decreasing the inter-granular pore-size $d_>$ with fixed $S_> = 100$ su. These results are consistent with the range $(T_1/T_2)_> \approx 3 \leftarrow 6$ previously observed for light producible hydrocarbons in shale, which we conclude is a function of the degree of confinement for the inter-granular (i.e. the more producible) hydrocarbons, together with diffusive-coupling effects with the dissolved (i.e. the less producible) hydrocarbons in kerogen.

From T_2 - T_2 exchange maps we find off-diagonal exchange peaks as the storage time is increased, which is a evidence for weak diffusive-coupling for T_2 . Our analysis shows that the characteristic exchange time for heptane to diffuse between inter-granular pores and the dissolved state is $\tau_{ex} \approx 15$ ms. Using the dimensionless constants $C_1 = T_{1\mu}/\tau_{ex}$ and $C_2 = T_{2\mu}/\tau_{ex}$, where $T_{1\mu}$ and $T_{2\mu}$ are the relaxation times of dissolved heptane, we confirm that T_1 is in the intermediate diffusive-coupling regime ($C_1 \approx 1$), while T_2 is in the weak diffusive-coupling regime ($C_2 \approx 0.02$). We also postulate how reservoir conditions would affect the results.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

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References

- [1] Ramakrishnan T, Fordham E, Venkataramanan L, Flaum M, Schwartz L. New interpretation methodology based on forward models for magnetic resonance in carbonates Soc. Petrophys. Well Log Analysts 1999 SPWLA-1999-MMM.
- [2] Zhang GQ, Hirasaki GJ, House WV. Internal field gradients in porous media. *Petrophysics* 2003;44(6):422–34.
- [3] Anand V, Hirasaki GJ. Diffusional coupling between micro and macroporosity for NMR relaxation in sandstones and grainstones. *Petrophysics* 2007;48(4):289–307.
- [4] Song Y-Q, Carneiro G, Schwartz LM, Johnson DL. Experimental identification of diffusive coupling using 2D NMR. *Phys Rev Lett* 2014;113:235503.
- [5] Mitchell J, Souza A, Fordham EJ, Boyd A. A finite element approach to forward modeling of nuclear magnetic resonance measurements in coupled pore systems. *J Chem Phys* 2019;50:154708.
- [6] Vinegar YOR, Reznick I, Gordin Y, Singer PM, Wang X, Chen Z, Nguyen SV, Li W, Bradley T, Hirasaki GJ, Lake LW, Feinstein S, Hatzor YH, Vinegar HJ. What happens to the petrophysical properties of a dual-porosity organic-rich chalk during early-stage organic maturation? Soc. Petrophys. Well Log Analysts 2020 SPWLA-5092.
- [7] Wang X, Singer PM, Chen Z, Hirasaki GJ, Vinegar HJ. A new method of estimating tortuosity and pore size in unconventional formations using NMR restricted diffusion measurements. Soc Petrophys Well Log Analysts 2020. SPWLA-5046.
- [8] Washburn KE, Callaghan PT. Tracking pore to pore exchange using relaxation exchange spectroscopy. *Phys Rev Lett* 2006;97:175502.
- [9] Montellhet L, Korb J-P, Mitchell J, McDonald PJ. Observation of exchange of micropore water in cement pastes by two-dimensional T_2 - T_2 nuclear magnetic resonance relaxometry. *Phys Rev E* 2006;74:061404.
- [10] Moldovan D, Fecete R, Demco DE, Culeaand E, Blümich B. Monte-carlo simulations of the two-dimensional NMR T_2 - T_2 exchange of fluids in porous media. *Diff Fund* 2009;36A(3):20.1–3.
- [11] Valori A, Rodin V, McDonald PJ. On the interpretation of 1H 2-dimensional NMR relaxation exchange spectra in cements: Is there exchange between pores with two characteristic sizes or Fe3+ concentrations? *Cem Concr Res* 2010;40:1375–7.
- [12] Landeghem MV, Haber A, D'Espinose de Lacaillerie J-B, Blümich B. Analysis of multisite 2D relaxation exchange NMR. *Concepts Magn Reson* 2010;36A(3):153–69.
- [13] Mitchell J, Gladden LF, Chandrasekera TC, Fordham EJ. Low-field permanent magnets for industrial process and quality control Proj. *Nucl Magn Reson Spect* 2014;76:1–60.
- [14] Fleury M, Soualem J. Quantitative analysis of diffusional pore coupling from T_2 -store- T_2 NMR experiments. *J Colloid Interface Sci* 2009;336:250–9.
- [15] Shikhov I, Li R, Arns CH. Relaxation and relaxation exchange NMR to characterise asphaltene adsorption and wettability dynamics in siliceous systems. *Fuel* 2018;220:692–705.
- [16] Washburn KE, Cheng Y. Detection of intermolecular homonuclear dipolar coupling in organic rich shale by transverse relaxation exchange. *J Magn Reson* 2017;278:18–24.
- [17] Singer PM, Chen Z, Hirasaki GJ. Fluid typing and pore size in organic shale using 2D NMR in saturated kerogen isolates. *Petrophysics* 2016;57(6):604–19.
- [18] Straley CA. mechanism for the temperature dependence of the surface relaxation rate in carbonates. Soc Core Analysts 2002:SCA2002-27.
- [19] Ozen AE, Sigal R. T_1/T_2 NMR surface relaxation ratio for hydrocarbons and brines in contact with mature organic-shale reservoir rocks. *Petrophysics* 2013;54(1):11–9.
- [20] Rylander E, Singer PM, Jiang T, Lewis RE, McLin R, Sinclair SM. NMR T_2 distributions in the Eagle Ford shale: Reflections on pore size. Soc Petrol Eng 2013. SPE-164554-MS.
- [21] Jiang T, Rylander E, Singer PM, Lewis RE, Sinclair SM. Integrated petrophysical interpretation of eagle ford shale with 1-D and 2-D nuclear magnetic resonance (NMR). Soc Petrophys Well Log Analysts; 2013. SPWLA-2013-LL.
- [22] Singer PM, Rylander E, Jiang T, McLin R, Lewis RE, Sinclair SM. 1D and 2D NMR core-log integration in organic shale. Soc Core Analysts 2013. SCA2013-18.
- [23] Washburn KE, Birdwell JE. Updated methodology for nuclear magnetic resonance characterization of shales. *J Magn Reson* 2013;233:17–28.
- [24] Kausik R, Fellah K, Rylander E, Singer PM, Lewis RE, Sinclair SM. NMR petrophysics for tight oil shale enabled by core resaturation Soc. Core Analysts 2014 2014-73..
- [25] Daigle H, Johnson A, Gips JP, Sharma M. Porosity evaluation of shales using NMR secular relaxation Unconv. Resources Tech Conf 2014. 1905272.
- [26] Washburn KE. Relaxation mechanisms and shales. *Concepts Magn Reson* 2014;43(3):57–78.
- [27] Korb J-P, Nicot B, Louis-Joseph A, Bubici S, Ferrante G. Dynamics and wettability of oil and water in oil shales. *J Phys Chem C* 2014;118(40):23212–8.
- [28] Nicot B, Vorapalawut N, Rousseau B, Madariaga LF, Hamon G, Korb J-P. Estimating saturations in organic shales using 2D NMR. *Petrophysics* 2015;57(1):19–29.
- [29] Lessenger M, Merkel R, Medina R, Ramakrishna S, Chen S, Balliet R, Xie H, Bhattad P, Carnerup A, Knackstedt M. Subsurface fluid characterization using downhole and core NMR T_1 - T_2 maps combined with pore-scale imaging techniques. Soc Petrophys Well Log Analysts 2015. SPWLA-2015-UUU.
- [30] Birdwell JE, Washburn KE. Multivariate analysis relating oil shale geochemical properties to NMR relaxometry. *Energy Fuels* 2015;29:2234–43.
- [31] Fleury M, Romero-Sarmiento M. Characterization of shales using T_1 - T_2 maps. *J Petrol Sci Eng* 2016;137:55–62.
- [32] Kausik R, Fellah K, Rylander E, Singer PM, Lewis RE, Sinclair SM. NMR relaxometry in shale and implications on logging. *Petrophysics* 2016;57(4):339–50.
- [33] Sun B, Yang E, Wang H, Seltzer SJ, Montoya V, Crowe J, Malizia T. Using NMR to characterize fluids in tight rock unconventional and shale formations. Soc Petrophys Well Log Analysts 2016. SPWLA-2016-PP.
- [34] Sondergeld C, Tinni A, Rai C, Besov A. NMR considerations in shale evaluation. Soc Petrophys Well Log Analysts 2016. SPWLA-2016-RRR.
- [35] Yang D, Kausik R. ^{23}Na and ^1H NMR relaxometry of shale at high magnetic field. *Energy Fuels* 2016;30(6):4509–19.
- [36] Zhang B, Daigle H. Nuclear magnetic resonance surface relaxation mechanisms of kerogen. *Geophysics* 2017;82(6):15–22.
- [37] Romero-Sarmiento M-F, Ramiro-Ramirez S, Berthe G, Fleury M, Littke R. Geochemical and petrophysical source rock characterization of the Vaca Muerta formation, Argentina: Implications for unconventional petroleum resource estimations Int. J Coal Geol 2017;184:27–41.
- [38] Chen Z, Singer PM, Hirasaki GJ. Effects of bitumen extraction on the 2D NMR response of saturated kerogen isolates. Soc Petrophys Well Log Analysts 2017. SPWLA-2017-S.
- [39] Chen Z, Singer PM, Kuang J, Vargas M, Hirasaki GJ. Effects of bitumen extraction on the 2D NMR response of saturated kerogen isolates. *Petrophysics* 2017;58(5):470–84.
- [40] Valori A, den Berg SV, Ali F, Abdallah W. Permeability estimation from NMR time dependent methane saturation monitoring in shales. *Energy Fuels* 2017;31:5913–25.
- [41] Thern H, Horch C, Stallmach F, Li B, Mezzatesta A, Zhang H, Arro R. Low-field NMR laboratory measurements of hydrocarbons confined in organic nanoporous media at various pressures. *Microporous Mesoporous Mater* 2018;269:21–5.
- [42] Nicot J-PKB, Jolivet I. Dynamics and wettability of petroleum fluids in shale oil probed by 2D T_1 - T_2 and fast field cycling NMR relaxation. *Microporous Mesoporous Mater* 2018;269:7–11.
- [43] Tandon S, Heidari Z. Improved analysis of NMR measurement in organic-rich mudrocks through quantifying hydrogen-kerogen interfacial relaxation mechanisms. Soc Petrophys Well Log Analysts 2019. SPWLA-2019-MMM.
- [44] Xie H, Gan Z. Investigation of physical properties of hydrocarbons in unconventional mudstones using two-dimensional NMR relaxometry. Soc Petrophys Well Log Analysts 2019. SPWLA-2019-ZZZ.
- [45] Song Y-Q, Kausik R. NMR application in unconventional shale reservoirs - new porous media research frontier Prog. Nucl Magn Reson Spect 2019;112–113:17–33.
- [46] Faux D, Kogon R, Bortolotti V, McDonald P. Advances in the interpretation of frequency-dependent nuclear magnetic resonance measurements from porous material. *Molecules* 2019;24:3688.
- [47] Ye S-J, Scribner A, McLendon D, Ijason O, Chen S, Shao W, Balliet R. Method of determining unconventional reservoir saturation with NMR logging. Soc Petrol Eng 2019. SPE-196069-MS.
- [48] Dang ST, Sondergeld CH, Rai CS. Interpretation of nuclear-magnetic resonance response to hydrocarbons: Application to miscible enhanced-oil-recovery experiments in shales. Soc Petrol Eng J 2019;SPE-191144-PA:302–9.

- [49] Kanwar JSS, Tutuncu AN, Yang Y. A high frequency NMR investigation of eagle ford shale cores. *Unconv Resour Tech Conf 2019*;URTeC-203-MS:1–14.
- [50] Zhu F, Hu W, Cao J, Ferrante G, Pasin M, Zhou B, Korb J-P. Probing dynamics and wettability of water and oil in conventional and unconventional sandstone rock cores by field-cycling NMR relaxometry. *Energy Fuels* 2019. <https://doi.org/10.1021/acs.energyfuels.9b02199>.
- [51] Wang H, Sun B, Yang Z, Seltzer SJ, Wigand MO. Accurate rock mineral characterization with nuclear magnetic resonance. *Unconv Resources Tech Conf 2019*:1–15. 2019-19-MS.
- [52] Tandon S, Heidari Z. Quantifying the mechanisms contributing to nuclear-magnetic-resonance surface relaxation of protons in kerogen pores of organic-rich mudrocks. *Soc Petrol Eng J* 2019;24(6). SPE-197063-PA.
- [53] Yang A, Firdaus G, Heidari Z. Electrical resistivity and chemical properties of kerogen isolated from organic-rich mudrocks. *Geophysics* 2016;81(6):643–55.
- [54] Craddock PR, Miles J, Lewis RE, Pomerantz AE. Thermal maturity-adjusted log interpretation (TMALI) in organic shales. *Petrophysics* 2019;60(5):540–59.
- [55] Singer PM, Chen Z, Alemany LB, Hirasaki GJ, Zhu K, Xie ZH, Vo TD. NMR relaxation of polymer-alkane mixes, a model system for crude oils. *Soc Petrophys Well Log Analysts* 2017. SPWLA-2017-XX.
- [56] Singer PM, Chen Z, Alemany LB, Hirasaki GJ, Zhu K, Xie ZH, Vo TD. Interpretation of NMR relaxation in bitumen and organic shale using polymer-heptane mixes. *Energy Fuels* 2018;32(2):1534–49.
- [57] Singer PM, Valiya Parambathu A, Wang X, Asthagiri D, Chapman WG, Hirasaki GJ, Fleury M. Elucidating the ^1H NMR relaxation mechanism in polydisperse polymers and bitumen using measurements, MD simulations, and models. *J Phys Chem B* 2020. <https://doi.org/10.1021/acs.jpcc.0c01941>.
- [58] Valiya Parambathu A, Singer PM, Hirasaki GJ, Chapman WG, Asthagiri D. Critical role of confinement in the NMR surface relaxation and diffusion of n-heptane in a polymer matrix revealed by MD simulations. *J Phys Chem B* 2020. <https://doi.org/10.1021/acs.jpcc.0c00711>.
- [59] Etminan SR, Javadpour F, Maini BB, Chen Z. Measurement of gas storage processes in shale and of the molecular diffusion coefficient in kerogen. *Int J Coal Geol* 2014;123:10–9.
- [60] Liu J, Chapman WG. Thermodynamic modeling of the equilibrium partitioning of hydrocarbons in nanoporous kerogen particles. *Energy Fuels* 2019;33(2):891–904.
- [61] Restuccia F, Ptak N, Rein G. Self-heating behavior and ignition of shale rock. *Combust Flame* 2017;176:213–9.
- [62] Larsen JW, Ashida R, Doetschman DC. Kerogen chemistry 6. involvement of native free radicals in kerogen thermal deoxygenation. *Phys Rev Lett* 2005;19:2216–7.
- [63] Acholla FV, Orr WL. Pyrite removal from kerogen without altering organic matter: The chromous chloride method. *Energy Fuels* 1993;7:406–10.
- [64] Venkataraman L, Song Y-Q, Hürlimann MD. Solving fredholm integrals of the first kind with tensor product structure in 2 and 2.5 dimensions. *IEEE Trans Sig Process* 2002;50(5):1017–26.
- [65] Song Y-Q, Venkataraman L, Hürlimann MD, Flaum M, Frulla P, Straley C. T_1 - T_2 correlation spectra obtained using fast two-dimensional laplace inversion. *J Magn Reson* 2002;154:261–8.
- [66] Brownstein KR, Tarr CE. Importance of classical diffusion in NMR studies of water in biological cells. *Phys Rev A* 1979;19(6):2446–53.
- [67] Yan W, Sun J, Golsanami N, Li M, Cui L, Dong H, Sun Y. Evaluation of wettabilities and pores in tight oil reservoirs by a new experimental design. *Fuel* 2019;252:272–80.
- [68] Newgord C, Tandon S, Heidari Z. Simultaneous assessment of wettability and water saturation using 2D NMR measurements. *Fuel* 2020;270:117431.
- [69] An Y, Althaus SM, Liu H-H, Chen J-H. Nuclear magnetic resonance measurement of methane diffusion in organic rich shales. *Fuel* 2019;247:160–3.
- [70] Lo S-W, Hirasaki GJ, House WV, Kobayashi R. Mixing rules and correlations of NMR relaxation time with viscosity, diffusivity, and gas/oil ratio of methane/hydro-carbon mixtures. *Soc Petrol Eng J* 2002;7(1):24–34.